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CASCADE: oracle-in-the-loop active learning for Cas13 mining and allosteric switch design

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Software

- Review
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Summary

CASCADE (Cas Collateral Activation — Discovery and Engineering) is open-source software for (1) mining Cas13-like effectors from metagenomic contigs with strict quality filters and (2) redesigning only inter-domain **linkers** so a structure model predicts clearer OFF versus ON catalytic geometry. Recognition (REC / native crRNA) and wild-type HEPN sequences stay frozen. PXDesign and ProteinMPNN propose linker changes; Protenix or Cattle-Prod score unbound, trigger-bound, and mismatch complexes; `EvolutionGym` converts relative fitness into a ProteinMPNN PSSM for the next generation. We demonstrate the stack on three verified *Listeria booriae* Cas13a proteins that underpin U.S. Provisional Patent Application No. 64/033,745 (Waite, 2026a), and shortlist linker morphs with positive composite fitness for continued evolution and wet-lab testing. All scores here are *in silico*.

Statement of need

Calling a structure model or inverse folder once is easy; running them as a **closed, constrained search** is not (Abramson et al., 2024; ByteDance AML AI4Science Team, 2024; Dauparas et al., 2022). Unconstrained redesign often damages catalysis, and single-state confidence does not measure a switch. CRISPR CAD tools such as ADAPT (Metsky et al., 2022) and BADGERS (Srinivas et al., 2024) optimize detection guides. CASCADE targets **effector mechanics**: freeze maps, OFF/ON/mismatch scoring, hierarchical oracle cost, and a miner that logs rejection reasons so false Cas13 calls do not poison search (Waite, 2026b). Users are computational biologists and engineers wrapping PXDesign (ByteDance, 2025), ProteinMPNN, and Protenix-compatible oracles (optional Cattle-Prod).

State of the field

Diffusion binders and global MPNN redesign optimize interfaces or monomer metrics. CASCADE freezes recognition and catalysis, mutates mechanical couplers, scores conformational hypotheses, stitches wild-type HEPNs after generation, and logs JSONL for optional post-training. Generators and oracles stay external; weights are not redistributed.

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Software design

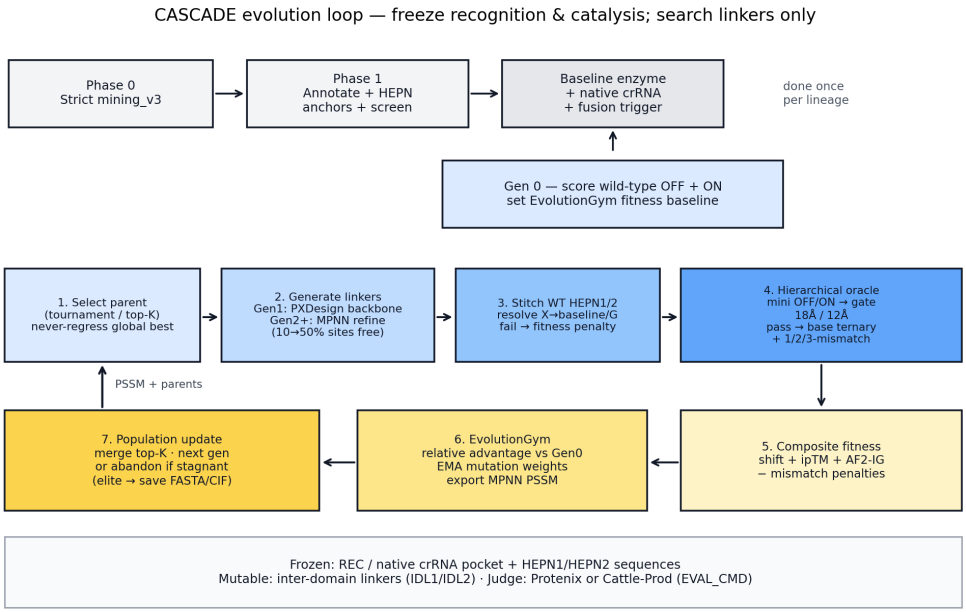


Figure 1: CASCADE loop: mine and bootstrap once, then select → generate linkers → stitch HEPNs → hierarchical OFF/ON/mismatch scoring → EvolutionGym PSSM → next generation.

Entrypoint: `scripts/evolution_orchestrator.py` (`scripts/smoke_gpu.sh` for a short GPU run).

- **Phase 0 — mining (`mining_v3`).** Length gates, canonical `R-X(4-6)-H` HEPN pairs, hard anti-signatures, reciprocal Cas13 homology, array-aware direct repeats, rejection reason per ORF.
- **Phase 1 — bootstrap.** SQLite annotation, HEPN anchors, native crRNA, optional structure screens.
- **Phase 2 — evolution.** Gen 0 scores wild type. Gen 1: PXDesign linker backbones. Gen 2+: ProteinMPNN on a cached backbone (10–50% linker sites free, bias-weighted). Stitch WT HEPNs; mini OFF/ON gates expensive ternary and mismatch jobs; relative fitness updates the PSSM.

Methods

Mining. ORFs (default 600–1400 aa) need a valid HEPN pair, no hard non-Cas13 signatures, and reciprocal hits to `data/cas13_reference_db.fasta` (remine defaults 30% identity / 80 aa). CRISPR DRs come from nearby arrays, not naïve k-mers (Waite, 2026b).

Freeze map. For the three *L. booriae* baselines, HEPN1 is residues 484–594 and HEPN2 is 795–905 (1-based). IDL1 sits before HEPN1; IDL2 between HEPNs. `hepn_structural_stitch.py` restores WT HEPNs after sampling; stitch failure costs a fixed fitness penalty (default 5.0).

Triggers and oracle. Fusion RNAs in `data/fusion_targets.json` build OFF (protein+crRNA) and ON (plus target) jobs; 1/2/3-nt spacer mismatches probe leak. `EVAL_CMD` selects Cattle-Prod or Protenix. His NE2–NE2 distance gates default to OFF 18 Å and ON 12 Å (**design thresholds**, not calibrated biophysics). Passers may receive base ternary evaluation.

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Generators and gym. Gen 1: PXDesign. Gen 2: ProteinMPNN with unfreeze `min(0.10+0.05·gen, 0.50)`, `--pssm_multi 0.5`, annealing temperature. Fitness rewards OFF−ON shift, ipTM, and AF2-IG, doubles for full ternary, and penalizes mismatch closure. EvolutionGym centers on Gen-0 fitness, EMA-updates substitution weights (`0.5/0.5`), clips to $[-5, 5]$, and exports `mpnn_bias_gen_*.json`. Elite requires ipTM 0.85, AF2-IG 0.80, and ON 12 Å. Seeds: `--seed` / `CASCADE_SEED`. CPU tests cover core helpers; figures rebuild via `scripts/build_paper_figures.py`.

Demonstration: three Cas13a baselines and morphs

Strict remine kept three *L. booriae* Cas13a ORFs with adjacent CRISPR arrays (Waite, 2026b):

Baseline (abbrev.)	Contig	aa	Native DR (truncated)	Best ref
NZ_JAASWF…_HIGH	NZ_JAASWF…	1064	GAUCAAAACAGAAGAGGACUA…	LseCas13a ~39%
NZ_JAARORO…_HIGH	NZ_JAARORO…	1056	GAUACCUCAAAACAGAAGAGGACUAA…	LseCas13a ~39%
NZ_JAARYF…_HIGH	NZ_JAARYF…	1052	GAUUAGAGUACCUCAAAACAGAAGAG…	LseCas13a ~39%

Campaign 2 (*Bacteroides* / *Flavobacterium* / *Leptotrichia*) accepted **0/102** ORFs under the same filters (Figure 2). These three scaffolds are the sequences covered by U.S. Provisional Patent Application No. 64/033,745 (Waite, 2026a); CASCADE is the discovery and design software behind that filing.

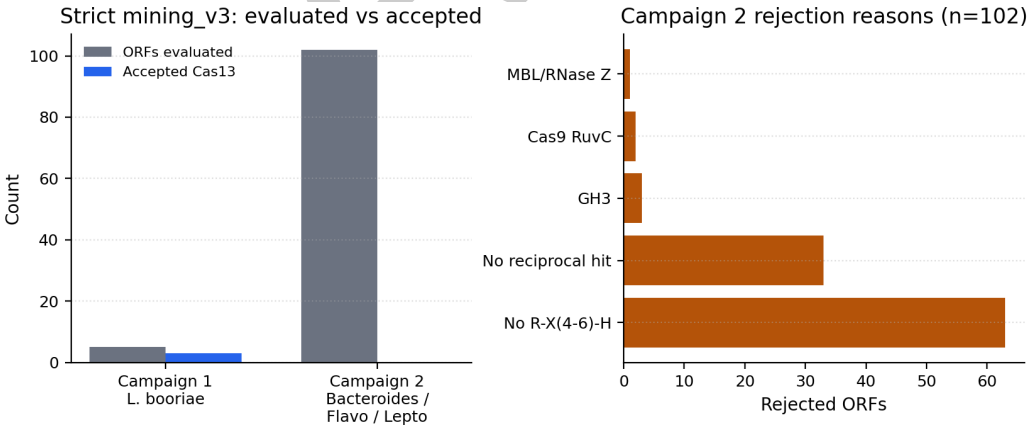


Figure 2: mining_v3 Campaign 1 vs Campaign 2: evaluated vs accepted ORFs, and Campaign 2 rejection reasons.

Intended use (unvalidated): fusion-RNA–gated collateral RNase (e.g. BCR-ABL1, EWSR1-FLI1) or Cas13 diagnostics. Search concentrated at the HEPN1 border (**475–484**) and **IDL2 ~700–780** (Figure 4).

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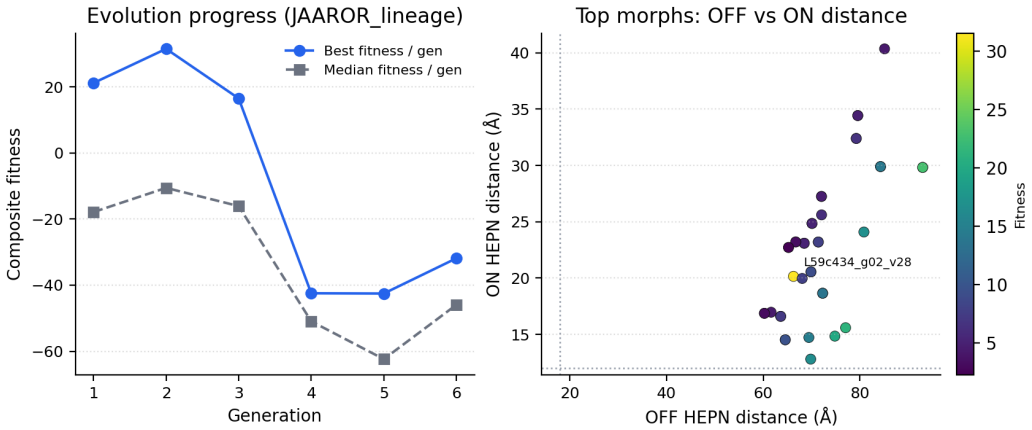


Figure 3: JAAROR: fitness by generation and OFF vs ON distances for top morphs (gates OFF 18 Å, ON 12 Å).

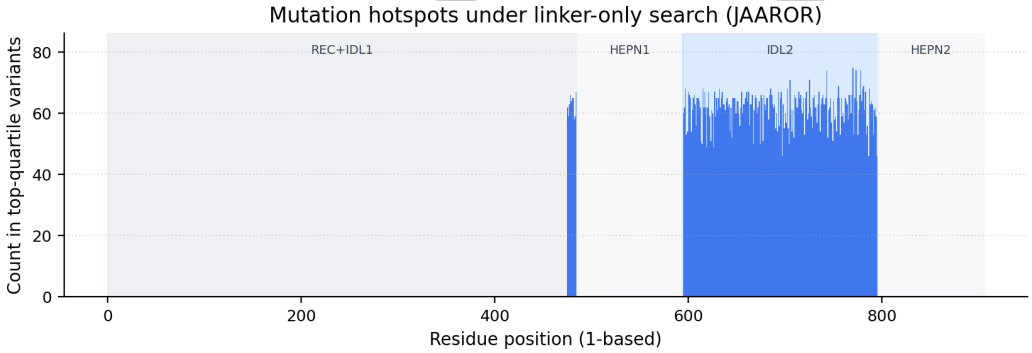


Figure 4: Top-quartile mutation hotspots on JAAROR; blue = IDL2.

No archived run met elite cuts under the limited GPU budget. Best JAAROR morph `L59c434_g02_v28` reached fitness 31.5 (OFF 66 Å, ON 20 Å, ipTM 0.34); neighbors with large OFF→ON shifts are in `paper/figures/wetlab_shortlist_JAAROR.csv`. Those candidates—and the three parental scaffolds—are still being evolved in follow-on CASCADE runs. Absence of an elite hit reflects incomplete search under funding constraints, not scaffold exhaustion.

Research impact statement

CASCADE is already the operational pipeline for the author's Cas13 switch program: it produced the three remine-verified *L. booriae* Cas13a baselines that are claimed in U.S. Provisional Patent Application No. 64/033,745 (Waite, 2026a), and it generated the JAAROR linker morphs with the strongest composite fitness in the archived campaign (e.g. `L59c434_g02_v28` and the wet-lab shortlist). Those morphs remain under active CASCADE evolution while GPU time allows. Reproducible mining tables, RL JSONL, figures, Apache-2.0 code, dual-use notes, CPU tests, and the Zenodo software archive (Waite, 2026b) document that research use for others to extend.

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AI usage disclosure

Coding assistants (including Cursor) helped with packaging and earlier manuscript drafts. Methods, figures, and claims were checked by the author against remine docs and RL logs. Assistants were not used in place of experiments. Core design decisions (freeze/evolve/score policy, gates, mining filters) were made by the author.

Acknowledgements

Built on Protenix, PXDesign, ProteinMPNN, and related open tools. No external grant funded the evolution campaigns reported here.

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